

Resolution Geometry

Residual Characteristic Geometry: Tame Reduction, the Hyperbolic Lock, Wild Certificates and Dynamic Cones

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Abstract

Rung L59 of the theorem ladder [1] left open the characteristic sets that are neither quadratic, Lorentz–Finsler, multicone nor stratified. This paper closes that rung, except for one class that it proves cannot be closed.

The residual splits along one dividing line: tameness. Characteristic sets definable in an o-minimal structure, which includes every set cut out by polynomial or globally analytic conditions, have:

- a finite canonical stratification;
- finitely many sheets, bounded uniformly in bounded-complexity families (Harnack bound in the algebraic case);
- integer dimension;
- a type given by finite combinatorial data;
- continuous moduli under linear equivalence, so they are classified up to stratified homeomorphism and not up to $GL(V)$.

Physically realized characteristic sets satisfy a stronger lock. If the principal symbol of the propagation law is hyperbolic, which for constant coefficients is equivalent to a well-posed Cauchy problem, then:

- the forward cone is convex;
- the sheets are real, ordered and nested, with no label monodromy;
- the only residual structure is singular contact between sheets, which carries $\mathbb{Z}_2$ polarization holonomy.

Conversely, a residual characteristic set of a principal symbol forces complex roots and Hadamard ill-posedness, an observable high-frequency instability.

The non-tame (wild) class contains fractal and infinitely accumulating direction sets. It is characterized by resolution invariants of its ε -components. No finite-resolution datum certifies tameness or wildness absolutely, so every certificate is indexed by its window. Smoothness and strict convexity are exchanged by duality. Characteristic cone fields of chaotic dynamics exist globally only under uniform hyperbolicity.

A ten-check battery verifies each statement on computed, label-free witnesses. It passes 10/10 on five seeds. A held-out certifier types 200 random instances of eight classes per seed with no wrong-type certificate on any seed (abstention 0.5–3%), and four mutation canaries are caught.

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1 The characteristic object

Let $V = T_q Q$ be the observable tangent space at a point of a regular stratum, and let $A \subset V \setminus \{0\}$ be a characteristic or admissibility set that is a natural module of the joint law. By natural descent it is defined on the quotient and invariant under the effective symmetry group (L13 of the ladder).

Definition 1.1 (Cone, body, projective set). A is a *cone* if $rA = A$ for all $r > 0$. Its projective set is $\mathbb{P}A = A/\mathbb{R}_{>0} \subset S(V)$. If membership depends on the radius, A is a *body*. Its Bouligand tangent cone at 0 and its recession cone are then separate natural modules, and only these are typed as cones.

Proposition 1.2 (Presentation invariance; RX01). *If $A = \{p \geq 0\}$, then for every positive function g and every strictly increasing odd φ , $\{\varphi(gp) \geq 0\} = A$ and $\{\varphi(gp) = 0\} = \{p = 0\}$. Degree, factorization and analytic presentation are therefore not invariants of $\mathbb{P}A$. On 40 random quartics with positive factors and $\varphi \in \{\sinh, z^3, \arctan 5z\}$, sign mismatches over 8×10^5 directions: 0.*

2 Observing a set at finite resolution

Definition 2.1 (ε -components). An observation of $\mathbb{P}A$ at angular resolution ε reports the $\varepsilon/2$ -thickening of $\mathbb{P}A$. Its number of connected components is $N_A(\varepsilon)$. On the projective circle, $N_A(\varepsilon)$ is the number of gaps of $\mathbb{P}A$ wider than ε .

Remark 2.2. A detector that reports a cell only above a measure threshold does not see sets of measure zero. Such sets then lie in the observational kernel of that detector, as the sensor-precision layer of the null (L26). The thickening model is the most sensitive finite-resolution observation and is used throughout.

Definition 2.3 (Window certificate). For resolutions $\varepsilon_1 > \dots > \varepsilon_m$, the window certificate is:

- STABLE if the last three counts agree;
- GROWTH if the log-count slope per factor of four in resolution exceeds 0.15 on the last three steps;

- ABSTAIN otherwise.

Proposition 2.4 (Window lemma; RX06). *Neither tameness nor wildness is certified by finite-resolution data. The certificate is a statement about its window only.*

- A tame cluster. *Thirty arcs whose gaps are log-uniform in $[2 \times 10^{-6}, 10^{-3}]$ give counts 3, 6, 16, 23 at $\varepsilon = 2\pi/4^6, \dots, 2\pi/4^9$, certified GROWTH. They stabilize at 32 by $\varepsilon = 2\pi/4^{11}$.*
- A late-wild set. *An arc plus a Cantor set of diameter 10^{-5} gives count 2 on every window down to $2\pi/4^{10}$, certified STABLE. Below that scale it grows (3, 5, 17).*

3 Tame reduction

Definition 3.1 (Tame). *A is tame if $\mathbb{P}A$ is definable in an o-minimal expansion of the real field. Examples are semialgebraic sets and globally subanalytic sets, which cover every characteristic set given by finitely many polynomial or analytic conditions on a compact parameter range.*

Imported Theorem 3.2 (Tame topology; [16, 11, 6]). *A definable set admits a finite Whitney stratification into connected definable manifolds. It has finitely many connected components, and its dimension is an integer equal to its box dimension. In a definable family, only finitely many topological types occur (Hardt triviality).*

Theorem 3.3 (Tame reduction). *Let A be a tame closed cone. Then:*

1. $N_A(\varepsilon)$ is eventually constant as $\varepsilon \rightarrow 0$, equal to the number of components of $\mathbb{P}A$;
2. $\mathbb{P}A$ has no infinite discrete subset and no fractal part; its box dimension is an integer;
3. $\mathbb{P}A$ carries a canonical coarsest Whitney stratification $\mathcal{S}$ whose boundary strata have constant rank and signature of the second fundamental form;
4. the tame type of A is the finite datum

$$\tau(A) = (\mathcal{S}, \text{ incidences, dim, signature on each stratum, } \\ \text{monodromy of sheets over the complement of the discriminant});$$

5. in a definable family of bounded complexity, finitely many tame types occur up to stratified homeomorphism.

Proof. (1) and (2) follow from finiteness of components and integrality of dimension (Imported Theorem 3.2): two components at positive distance are separated once ε is below that distance. (3) The loci where the rank or signature of the second fundamental form changes are definable, so they refine $\mathcal{S}$ finitely; the coarsest such refinement is canonical. (4) collects the data of (3). (5) is Hardt triviality applied to the family. $\square$

Corollary 3.4. *The quadratic, Lorentz–Finsler, multicone and stratified branches are particular tame types. The residual tame cones are the other values of τ : cornered convex, weakly convex with flat faces, connected nonconvex, and branch-monodromic. Each is a finite combinatorial datum. The residual of L59 that is not a tame type is exactly the wild class of Section 6.*

Proposition 3.5 (Algebraic sheet bound; RX02; [7]). *If $\mathbb{P}A$ is the real zero set of a degree- d curve in $\mathbb{RP}^2$, the number of sheets is at most $(d-1)(d-2)/2 + 1$. For 40 random Kostlan quartics, oval counts were stable across the two finest of three resolutions in 100% of cases. The maximum was 2 ovals (bound 4); the histogram was $34 \times 1, 6 \times 2$.*

No-Go 3.6 (No finite linear classification; RX03). Already polyhedral cones in $\mathbb{R}^3$ with five extreme rays have two continuous projective invariants of the ray configuration. Two random convex pentagonal cones differ in them by 0.33, the invariants are exact under $GL(3)$ (error 0), and along a one-parameter family that stays in convex position they move monotonically and continuously (total change 0.078, maximal step 0.0023 over 40 steps). Tame types are classified up to stratified homeomorphism, not up to linear equivalence.

4 Duality

Imported Theorem 4.1 (Smoothness and strict convexity are dual; [14, 15]). *A closed convex pointed cone with nonempty interior is smooth (one supporting hyperplane at each boundary ray) if and only if its dual cone is strictly convex (no segment in its boundary). The converse also holds.*

Proposition 4.2 (Corners and flats; RX07). *Corners are certified when the maximal derivative jump of the radial function does not decay under resolution escalation; flats are certified by zero curvature on a set of positive measure.*

- Stadium section. *No corners, flats on 44% of the angle.*
- Its polar. *Corners, no flats.*
- An ellipse. *Neither.*

A Lorentz–Finsler realization requires a smooth, strictly convex section, that is, a primal and a dual that are both smooth.

5 The hyperbolic lock

Definition 5.1 (Hyperbolic symbol). A homogeneous polynomial p on V is *hyperbolic* in direction e if $p(e) \neq 0$ and $s \mapsto p(v - se)$ has only real roots for every v . Its *hyperbolicity cone* $\Lambda(p, e)$ is the component of $\{p \neq 0\}$ containing e .

Imported Theorem 5.2 (Gårding; Renegar; Hadamard; [4, 5, 9, 13]). *1. $\Lambda(p, e)$ is an open convex cone, and p is hyperbolic in every direction of it.*

- 2. For constant-coefficient operators, the Cauchy problem is well posed for all lower-order terms iff the principal symbol is hyperbolic. The support of the fundamental solution lies in the dual of the closed hyperbolicity cone.*
- 3. The derivative $D_e p$ is hyperbolic, with $\Lambda(p, e) \subseteq \Lambda(D_e p, e)$.*
- 4. In three variables the real projective curve of a hyperbolic p of degree d consists of $\lfloor d/2 \rfloor$ nested ovals around $[e]$, plus a pseudo-line if d is odd [8].*

Theorem 5.3 (Hyperbolic lock). *Let A be the characteristic set of a propagation law whose principal symbol p is hyperbolic in a time direction e . Then:*

- 1. the forward cone is convex;*
- 2. the sheets of $\mathbb{P}A$ are real, ordered and nested, so sheet labels have trivial monodromy around every loop that avoids sheet contact;*
- 3. the only residual structure is contact between sheets. Around an isolated conical contact the polarization (eigenvector) line bundle has holonomy -1 ;*
- 4. connected nonconvex forward cones, wild sets and label-exchanging sheets do not occur.*

Proof. (1) is Imported Theorem 5.2(1). (2) Along every line through e the roots are real and can be ordered; ordering is continuous, so a loop that avoids coincidences returns each sheet to itself. Nestedness is (4) in three variables and the interlacing of real roots in general. (3) Near a conical contact of a symmetric pencil the two eigenvectors form the tautological real line bundle over the loop, which is the Möbius bundle. (4) A nonconvex forward cone contradicts (1). A wild set contradicts the semialgebraicity of $\{p = 0\}$ (Theorem 3.3). Label exchange contradicts (2). $\square$

Proposition 5.4 (Checks; RX04, RX05). • RX04. For D_e^j of a product of six random linear forms on $\mathbb{R}^4$ ($j = 1, 2$; curved, not polyhedral cones): complex roots on 0/400 random lines; midpoint-convexity violations 0/300; nesting violations $\Lambda_0 \subseteq \Lambda_1 \subseteq \Lambda_2$: 0.

- RX05. For a random 3×3 symmetric pencil the conical point was located to gap $< 10^{-12}$. Around it the minimal eigenvalue gap stayed positive (1.0×10^{-3}), the sheets were not exchanged, and the polarization holonomy was exactly -1 .

Theorem 5.5 (Residual means ill-posed; RX10). If the characteristic set of a principal symbol is residual in any sense excluded by Theorem 5.3, then p is not hyperbolic. There are then real covectors ξ with complex roots $\tau(\xi)$, and plane waves grow like $\exp(s |\operatorname{Im} \tau(\xi)| t)$ at frequency s . The growth is unbounded in frequency, so the Cauchy problem is ill posed.

Proof. Contrapositive of Theorem 5.3 and Imported Theorem 5.2(2). The linear scaling follows from homogeneity, $\tau(s\xi) = s\tau(\xi)$. $\square$

For real coefficients, monodromy commutes with complex conjugation: label exchange always exchanges conjugate pairs together. In RX05, the non-hyperbolic quartic with roots $\pm i \pm \sqrt{z}/2$ gives the permutation (12)(34) around $z = 0$. In RX10, a perturbation of a hyperbolic quartic has complex roots on 96% of lines, with growth rate per unit frequency 0.0485 at $s = 1, 10, 100$ (constant to all digits). The unperturbed symbol has none.

Remark 5.6. In RG terms, Theorem 5.5 makes the residual observable. An experiment whose characteristic module is residual shows unbounded amplification of high-frequency perturbations, which is a falsifiable signature on the joint law.

6 The wild class

Definition 6.1 (Wild). A is wild if $\mathbb{P}A$ is not definable in any o-minimal structure. Its resolution invariants are the count exponent $\delta_A = \limsup_{\varepsilon \rightarrow 0} \log N_A(\varepsilon) / \log(1/\varepsilon)$, the box dimension, and the Cantor–Bendixson rank of its accumulation set.

Proposition 6.2 (Wild witnesses). • Ternary Cantor direction set. $\delta = \log 2 / \log 3$, box dimension $\log 2 / \log 3$, Cantor–Bendixson rank ∞ (perfect).

- Arcs of width w/j^2 at angles c/j . $\delta = 1/2$, box dimension $1/2$, one accumulation point (rank 2).

Neither is tame, since $N_A(\varepsilon)$ is unbounded (Theorem 3.3(1)).

No-Go 6.3 (No finite classification of the wild class). Closed subsets of the circle up to homeomorphism already include all countable compacta. These are classified by Cantor–Bendixson rank, an arbitrary countable ordinal [12]. No finite combinatorial type exists, and by Proposition 2.4 finite data certify only window statements.

7 Typed certifier

The certifier reads only sampled directions.

Single-sheet sections. The input is the radial function at angular resolutions $2^9, 2^{12}, 2^{15}$.

- *Corner*: the ratio of the maximal derivative jump at the finest to the coarsest level is ≥ 0.7 at both 2^{15} and 2^{17} ; smooth if it is ≤ 0.35 ; otherwise ABSTAIN.
- *QUADRATIC*: quadratic-fit residual of ρ^{-2} below 10^{-6} at two levels.
- *FINSLER*: smooth, with normalized polar curvature $\rho^2 + 2\rho'^2 - \rho\rho'' > 0.05$ everywhere.
- *NONCONVEX*: curvature < -0.05 somewhere.
- *WEAKLY-CONVEX*: smooth, with flats on more than 5% of the angle.
- *CORNERED-CONVEX*: corners, and no negative curvature away from them.

Sheet sets. The input is $N_A(\varepsilon)$ at $\varepsilon = 2\pi/2^k$, $k = 10, 12, \dots, 24$, with the window certificate.

Proposition 7.1 (Held-out validation; RX09). *Twenty-five random instances were drawn per class, with no parameters or labels seen by the certifier:*

<i>class (generator)</i>	<i>certificate</i>	<i>seed 20261001</i>
<i>quadratic (random ellipse)</i>	<i>QUADRATIC</i>	<i>25/25</i>
<i>Finsler (support $1 + a \cos 3\phi + b \cos 2\phi$)</i>	<i>FINSLER</i>	<i>25/25</i>
<i>convex polygon</i>	<i>CORNERED-CONVEX</i>	<i>24/25 (1 abstention)</i>
<i>star-shaped $1 + a \cos k\theta$, negative curvature</i>	<i>NONCONVEX</i>	<i>25/25</i>
<i>polygon $\oplus$ disk of radius 10^{-3}</i>	<i>WEAKLY-CONVEX</i>	<i>25/25</i>
<i>finite arcs</i>	<i>STABLE</i>	<i>25/25</i>
<i>finite fine arcs (widths $2^{-5} \times 10^{-5}$)</i>	<i>STABLE</i>	<i>25/25</i>
<i>Cantor or accumulation</i>	<i>GROWTH</i>	<i>25/25</i>

Across seeds 20261001, 1, 42, 777 and 2026 the wrong-type rate is 0 on every seed, with abstention 0.5%, 1.5%, 2%, 2.5% and 3%. Abstentions occur on polygons and rounded polygons whose derivative-jump ratio leaves the undecided band only beyond the window, which is the resolution limit of Proposition 2.4 for corners. A one-step corner test misread up to 6/25 rounded polygons as cornered, and a single resolution misreads all of them (canary).

8 Dynamic characteristic cones

In a dynamical experiment the characteristic object is a field $x \mapsto C_x$ transported by the dynamics.

Imported Theorem 8.1 (Cone-field criterion; [3, 18, 10]). *A diffeomorphism is uniformly hyperbolic on a compact invariant set iff there is a continuous cone field with $Df_x C_x \subset \text{int } C_{f(x)}$ and uniform expansion on it. In that case the angle between the stable and unstable directions is bounded away from zero.*

Proposition 8.2 (Uniform versus nonuniform; RX08). *For the cat map $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ the splitting is constant (symmetric matrix, orthogonal eigenvectors, 90°), and the constant cone field certifies uniform hyperbolicity. For the Hénon map (1.4, 0.3) the minimal angle along the orbit is 0.63° , 0.054° and 0.0061° for orbit lengths 2×10^3 , 2×10^4 , 2×10^5 . It has no positive lower bound, so there is no continuous invariant cone field and the certificate is NONUNIFORM. Without length escalation the decrease cannot be seen (canary).*

Remark 8.3. For nonuniformly hyperbolic or chaotic experiments the characteristic process is typed per window, together with the equivariance $Df C_x = C_{f(x)}$ and the kernel and stabilizer conditions of [2]. Its tangencies are discriminant walls in time.

9 Ladder insertion

L59 is replaced by the following rungs, all dependent on L35 (realization branches) and L15 (strata).

Rung	Statement	Status	Ref.
L59.1	Bodies versus cones; tangent and recession cones are separate modules	forced	Def. 1.1
L59.2	Presentation invariance of $\mathbb{P}A$	forced	Prop. 1.2
L59.3	Finite-resolution observation by ε -components; window lemma	forced (no-go)	Prop. 2.4
L59.4	Tame reduction: finite type, finitely many sheets, integer dimension, Hardt finiteness	imported / forced	Thm. 3.3
L59.5	No finite linear classification	no-go	No-Go 3.6
L59.6	Duality of smoothness and strict convexity	imported	Thm. 4.1
L59.7	Hyperbolic lock: convex, ordered, nested; $\mathbb{Z}_2$ polarization holonomy at contacts	imported / forced	Thm. 5.3
L59.8	Residual characteristic symbol implies Hadamard ill-posedness	forced	Thm. 5.5
L59.9	Wild class: resolution invariants; no finite classification	no-go	Sec. 6
L59.10	Dynamic cones: global only under uniform hyperbolicity; otherwise window-typed	imported	Thm. 8.1

Open Problem 9.1. Enumerate the tame types of degree- d characteristic cones in dimension $1+n$. For $n = 2$ this contains the topological classification of real plane curves (Hilbert’s sixteenth problem, part one), which is known for $d \leq 7$ [17]. This is a problem of real algebraic geometry, not of RG.

10 Non-claims

- No physical causal cone is inferred from a wild or residual set. Theorem 5.5 states that such sets signal ill-posed propagation, not new propagation.
- The tame type is a classification up to stratified homeomorphism, not a complete invariant up to linear maps.
- The battery checks the stated witnesses and the certifier implementation. The theorems rest on the cited results and the proofs given.

A Reproducibility

`rg_residual_cone_attack.py` passes 10/10 on seeds 20261001, 1, 42, 777 and 2026. `--canary` removes, in turn:

- the escalation beyond the coarse window (RX06);
- length escalation in the dynamic cone test (RX08);
- resolution escalation in the corner test (RX09);
- the non-hyperbolic perturbation (RX10).

Each targeted check then fails.

Corrections made before registration are recorded in the code:

1. A measure-zero Cantor set is invisible to point sampling. Observation is modelled by ε -components.
2. The non-hyperbolic monodromy was first expected to be a single transposition. Conjugation symmetry forces (12)(34), and the gate was re-registered.
3. A single resolution sufficed on the first instance set, so a class whose type requires escalation (rounded polygons) was added.
4. The pentagon generator first allowed non-convex position; it was fixed.
5. A single corner threshold misread 6/25 rounded polygons on seed 777 and a one-step lock 2/25; corners now survive two escalations with an undecided band.
6. The moduli path first interpolated between two cones and left convex position; it now stays convex.
7. The Hadamard scaling test first used a covector with real roots; it now uses one with complex roots.

B Disposition of the earlier witnesses RC01–RC10

ID	Defect	Now
RC01	—	Definition 1.1
RC02	component count hard-coded	NONCONVEX class, computed (RX09)
RC03	corner rule fires on every curve (7 “corners” on an ellipse)	escalation corner test (RX07, RX09)
RC04	branch exchange by construction	RX05: monodromy implies non-hyperbolic
RC05, RC06	illustrations of known sets	ε -components, window lemma (RX06, RX09)
RC07	gate true for any two wedges; primal = dual	duality theorem (RX07)
RC08	topology hard-coded	stabilizer and topology are data of $\tau(A)$
RC09	thresholds tuned to agree	RX01
RC10	class index used as a feature	held-out certifier (RX09)

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